

Fitness Buddy

Problem Statement No. 13

The Challenge

In today's fast-paced world, many individuals struggle to maintain a healthy lifestyle due to lack of personalized guidance, time constraints, and inconsistent motivation. Traditional fitness solutions often require expensive subscriptions, in-person consultations, or rigid schedules that do not adapt to personal preferences or daily routines.

There is a growing need for an accessible, intelligent virtual assistant that can provide on-demand fitness advice, healthy lifestyle suggestions, and nutrition guidance tailored to individual needs.

Fitness Buddy aims to:

- Recommend personalized home workouts.
- Provide motivational tips and daily inspiration.
- Suggest healthy meal ideas.
- Encourage healthy habit-building and consistency.

Technology Requirement: Use IBM Cloud Lite Services and IBM watsonx.ai Granite models.

Proposed Solution

Fitness Buddy is an AI-powered virtual fitness coach built using IBM watsonx.ai and IBM Granite Guardian 3.8B. The application provides personalized workout plans, nutrition guidance, daily motivation, and progress tracking through an intelligent conversational interface. A multi-agent approach is used with Workout, Nutrition, Motivation, and Progress agents working together to deliver personalized recommendations. The system includes an interactive dashboard for tracking BMI, workouts, calories, and fitness progress, helping users maintain a healthy lifestyle.

Technologies Used

- IBM Bee/Bob
- IBM watsonx.ai
- IBM Granite Guardian 3.8B
- IBM Cloud Lite Services
- Python
- Flask
- HTML, CSS, JavaScript, Bootstrap
- SQLite
- Chart.js
- Git & GitHub